PHYSIOLOGY · MCQ PRACTICE

Physiology of the Defecation Reflex — MCQ Practice

LG 3 · Dr. Nazleen Shakir Akreyi
GIT BLOCK

Physiology GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Questions

1. What stimulates the defecation reflex? EASY

A. Contraction of the sigmoid colon
B. Distension of the rectal wall by fecal material
C. Gastric distension
D. Bile release into the duodenum
E. Ileocecal valve opening

2. Which anal sphincter is under voluntary control? EASY

A. Internal anal sphincter
B. External anal sphincter
C. Both are voluntary
D. Neither is voluntary
E. The ileocecal sphincter

3. Which nerve provides voluntary motor control to the external anal sphincter? EASY

A. Vagus nerve
B. Pelvic splanchnic nerve
C. Pudendal nerve
D. Hypogastric nerve
E. Phrenic nerve

4. Why can infants not voluntarily control defecation? EASY

A. They lack rectal stretch receptors
B. They lack an internal anal sphincter
C. Voluntary control of the external anal sphincter has not yet developed
D. Their sacral spinal cord is absent
E. Their myenteric plexus has not yet formed

5. A patient with a complete spinal cord transection at T10 (above the sacral segments) still has reflexive bowel emptying but cannot sense the urge or voluntarily delay it. Which component of the defecation reflex best explains the preserved reflexive emptying? MEDIUM

A. Cortical voluntary control, which remains fully intact
B. The intrinsic (myenteric) and sacral parasympathetic reflex arcs, which do not require intact cortical connections
C. The pudendal nerve, now hyperactive
D. Loss of all rectal sensation, which paradoxically increases reflex strength
E. Sympathetic outflow, now unopposed

6. A patient habitually ignores the urge to defecate for hours at a time. What happens to the fecal material during that delay? MEDIUM

A. It is expelled regardless of voluntary effort

B. It backs up into the sigmoid colon until the next mass peristaltic wave re-stimulates the stretch receptors

C. The rectum permanently loses its ability to sense distension

D. The internal anal sphincter remains open throughout

E. Defecation reflex sensitivity increases permanently with each postponement

7. During a normal defecation reflex, which correctly orders the sequence of events after rectal distension? MEDIUM

A. External sphincter relaxes first, then stretch receptors fire

B. Stretch receptors fire → sacral cord relays motor impulses → longitudinal rectal muscle contracts raising pressure → internal sphincter opens → external sphincter determines outcome

C. Internal sphincter opens before any sensory signal reaches the spinal cord

D. The rectum shortens before the sensory signal is even generated

E. Voluntary control always precedes the intrinsic reflex

8. A patient reports a bowel movement frequency of 3 times per week, with no other GI symptoms. How should this be interpreted physiologically? MEDIUM

A. This is pathological and always indicates constipation

B. This falls within the wide range of normal bowel habit (2–3/day to 3–4/week)

C. This indicates loss of the intrinsic myenteric reflex

D. This is only normal in infants

E. This indicates external anal sphincter dysfunction

9. Which statement correctly distinguishes the intrinsic (myenteric) component of the defecation reflex from the parasympathetic (sacral) component? MEDIUM

A. The intrinsic component is a local reflex mediated by the myenteric plexus, while the parasympathetic component originates from sacral segments S2–S4 and amplifies the response

B. The intrinsic component requires the pudendal nerve; the parasympathetic component does not

C. Both components are identical and interchangeable

D. The parasympathetic component is entirely voluntary

E. The intrinsic component is mediated by the vagus nerve

10. A patient voluntarily performs a Valsalva maneuver (forced expiration against a closed glottis) while attempting to defecate. What is the physiological rationale for this maneuver? MEDIUM

A. It relaxes the internal anal sphincter directly via the vagus

B. It increases intra-abdominal pressure, assisting rectal emptying alongside the reflex

C. It stimulates additional stretch receptors in the rectum

D. It has no physiological effect on defecation

E. It closes the external anal sphincter to prevent leakage

11. A physician examines a patient with suspected sacral nerve root damage (S2–S4). Which finding would most specifically support this diagnosis? MEDIUM

A. Loss of taste on the tongue

B. Impaired parasympathetic amplification of the rectal contraction/sphincter relaxation reflex, though the local myenteric reflex may partially persist

C. Loss of swallowing entirely

D. Loss of gastric acid secretion

E. Loss of pancreatic enzyme secretion

12. Which muscle, in addition to the internal anal sphincter, actively contracts to help maintain continence between defecation events? MEDIUM

A. Diaphragm

B. Puborectalis muscle

C. Masseter

D. External oblique only

E. Cricopharyngeus

13. A patient with a lesion affecting only the pudendal nerve (sparing the pelvic parasympathetic and sacral cord pathways) would most likely present with: MEDIUM

A. Complete loss of any rectal sensation

B. Loss of voluntary control over the external anal sphincter, with the reflexive components of defecation still present

C. Loss of the intrinsic myenteric reflex

D. Loss of internal anal sphincter tone specifically

E. No clinical effect at all

14. Sensory input from the anal canal skin, unlike input from the rectum itself, is carried by which type of afferent system? MEDIUM

A. Purely autonomic afferents to the enteric nervous system, with no cortical input

B. Somatosensory afferents to the CNS, carrying pain, temperature, and touch

C. Only stretch receptors identical to those in the rectal wall

D. No sensory innervation exists in the anal canal skin

E. Only chemoreceptor input

15. A patient has a complete cauda equina lesion (affecting sacral nerve roots bilaterally) after trauma. Predict the most accurate combined effect on defecation. HARD

A. Total abolition of any rectal emptying, since all reflex components are lost

B. Loss of the parasympathetic-amplified reflex and voluntary (pudendal-mediated) sphincter control, but the local intrinsic myenteric reflex may still allow some slow, uncoordinated rectal emptying

C. Normal defecation, since the cauda equina plays no role

D. Selective loss of only the internal sphincter, with everything else intact

E. Selective loss of only voluntary control, with the parasympathetic reflex fully intact

16. A physiologist proposes that defecation is "purely a spinal reflex requiring no higher input." Which clinical observation best refutes this claim? HARD

A. Infants defecate reflexively without cortical input, proving the claim true

B. Adults can voluntarily suppress or trigger defecation via cortical decisions that override or permit the sphincter response, showing cortical modulation is a functional part of the system

C. The intrinsic myenteric reflex operates without any spinal involvement

D. Sacral cord lesions have no effect on defecation

E. Pudendal nerve block has no effect on continence

17. A researcher measures rectal pressure and finds that voluntary contraction of the external anal sphincter during an active defecation reflex halts expulsion despite ongoing internal sphincter relaxation and rectal contraction. What does this demonstrate about the hierarchy of anal continence control? HARD

A. The internal sphincter is the final determinant of continence

B. Voluntary (external sphincter) control can override the involuntary components of the reflex, making it the final checkpoint before expulsion

C. The intrinsic myenteric reflex is the final checkpoint

D. Rectal pressure alone determines whether expulsion occurs, independent of any sphincter

E. The parasympathetic reflex overrides voluntary control

18. A patient with chronic diabetic autonomic neuropathy has selective damage to autonomic (parasympathetic and sympathetic) fibers, with somatic nerves relatively spared. Which defecation-related deficit is most specifically predicted? HARD

A. Loss of voluntary external sphincter control entirely

B. Impaired parasympathetic amplification of rectal contraction/internal sphincter relaxation, with voluntary pudendal-mediated sphincter control relatively preserved

C. Complete loss of anal canal skin sensation

D. Loss of the puborectalis muscle function entirely

E. No effect on defecation, since it is purely a somatic process

19. A comparison of the intrinsic myenteric reflex and the parasympathetic sacral reflex shows that lesioning the myenteric plexus locally (e.g., in Hirschsprung-like aganglionosis of the rectum) impairs defecation even when sacral nerve roots are fully intact. What does this best illustrate? HARD

A. The intrinsic and extrinsic (parasympathetic) reflex pathways are fully redundant, so either alone is always sufficient for normal defecation

B. The local myenteric circuit is a necessary component of the full reflex, and intact extrinsic (sacral) innervation alone cannot fully substitute for it

C. Sacral innervation is irrelevant to defecation

D. Myenteric plexus function is irrelevant if sacral roots are intact

E. This scenario would have no measurable effect on defecation

20. A patient can voluntarily initiate defecation at will even without prior rectal distension by bearing down forcefully. Which explanation best accounts for this? HARD

A. Voluntary abdominal/diaphragmatic contraction (Valsalva) combined with willful relaxation of the external sphincter can raise intrarectal pressure and open the internal sphincter mechanically, bypassing the need for reflex-triggering distension first

B. This is physiologically impossible; defecation always requires prior rectal distension

C. This proves the internal sphincter is entirely under voluntary control

D. This indicates a pathological reflex

E. The stretch receptors are being stimulated by the abdominal muscles directly

✓ Answer key

Q	Answer	Difficulty	Why
1	**B**	EASY	Rectal wall distension stimulates stretch receptors that initiate the defecation reflex.
2	**B**	EASY	The external anal sphincter (skeletal muscle) is voluntary, unlike the internal sphincter.
3	**C**	EASY	The pudendal nerve supplies voluntary (skeletal muscle) control of the external anal sphincter.
4	**C**	EASY	The reflex itself is present at birth; only voluntary external sphincter control develops later.
5	**B**	MEDIUM	The intrinsic enteric and sacral parasympathetic (S2–S4) reflex components can drive defecation locally without requiring communication with the cortex; only voluntary/conscious control is lost when that pathway is disrupted.
6	**B**	MEDIUM	Postponing defecation doesn't cancel the reflex — feces retreat into the sigmoid colon until re-stimulated by the next mass peristalsis wave.
7	**B**	MEDIUM	This is the correct stepwise sequence: sensory afferents → sacral cord → motor efferents → rectal contraction/pressure rise → internal sphincter relaxation → voluntary external sphincter decision.
8	**B**	MEDIUM	Normal bowel frequency ranges widely, from 2–3 times per day to 3–4 times per week.
9	**A**	MEDIUM	The intrinsic reflex is local (myenteric plexus); the parasympathetic reflex arises from sacral segments S2–S4 and amplifies rectal contraction and sphincter relaxation.
10	**B**	MEDIUM	Central/cortical control includes conscious relaxation of the external sphincter plus increased intra-abdominal pressure (Valsalva) to assist expulsion.
11	**B**	MEDIUM	S2–S4 carries the parasympathetic component of the defecation reflex; its loss impairs amplification of rectal contraction and sphincter relaxation, though the local intrinsic reflex can persist independently.
12	**B**	MEDIUM	Contraction of the internal anal sphincter together with the puborectalis muscle maintains continence.
13	**B**	MEDIUM	

			The pudendal nerve provides voluntary somatic control to the external anal sphincter; its isolated loss impairs voluntary continence/defecation control while the involuntary intrinsic and parasympathetic reflex components remain.
14	**B**	MEDIUM	The anal canal skin is innervated by somatosensory nerves carrying pain, temperature, and touch to the CNS, distinct from the rectal mechanoreceptors that supply the ENS.
15	**B**	HARD	A cauda equina lesion disrupts both the sacral parasympathetic outflow (S2–S4) and the pudendal nerve (also sacral), removing amplification and voluntary control, but the intrinsic myenteric reflex is a local gut wall circuit that can partially persist independent of extrinsic innervation.
16	**B**	HARD	The ability of adults to consciously permit or withhold defecation (central/cortical control deciding external sphincter relaxation and Valsalva) demonstrates that higher centres actively modulate the reflex, not merely observe it.
17	**B**	HARD	Even with the involuntary reflex fully active (rectal contraction, internal sphincter relaxation), voluntary contraction of the external anal sphincter can still prevent expulsion, showing it functions as the final voluntary checkpoint.
18	**B**	HARD	Autonomic (parasympathetic S2–S4) neuropathy would impair the amplifying reflex arc while the pudendal nerve, a somatic nerve, would be relatively spared, preserving voluntary sphincter control despite reflex weakness.
19	**B**	HARD	Local myenteric plexus disease (as in aganglionic segments) impairs the reflex even with intact sacral parasympathetic supply, showing that the intrinsic local circuit provides essential amplification/coordination the extrinsic reflex alone cannot fully replace.
20	**A**	HARD	Cortical/central control can drive voluntary Valsalva and external sphincter relaxation, mechanically raising intrarectal pressure enough to open the internal sphincter and initiate expulsion even without the reflex having been triggered by spontaneous distension first.